

Higgs Vacuum Route Status: Metastability, Lower Branches, and Standing Closure

Paper 4 of the Higgs Structural Endpoint Arc

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Abstract

This manuscript proves the fourth paper in the Higgs structural endpoint arc. Papers 1–3 fix the Higgs bridge/fixation carrier $C_H := H_{\text{br/fix}}$, identify Higgs structural identity as admissible occupation of C_H , and delete primitive Higgs-scale standing independent of that carrier. The present paper classifies Higgs-vacuum and vacuum-stability claims downstream of those results. Its target is not the numerical selection of Standard Model stability or metastability. Its target is the standing status of vacuum descriptions, lower branches, lifetime routes, thermal routes, and electroweak-baryogenesis-style routes once Higgs identity and scale standing are already carrier-indexed.

The paper distinguishes vacuum description, route availability, and standing closure. A vacuum location, effective-potential profile, lower branch, instability crossing, tunneling calculation, lifetime estimate, thermal transition, or electroweak-baryogenesis route may be physically meaningful and computationally important. None of these is a primitive Higgs ground. Each is a representative, route marker, route-available protocol, certificate-closed downstream status, target update, or failed primitive assertion. The vacuum route normal form is

$$R_H^{\text{vac}} = (D_H, C_H, R_H^{\text{scale}}, V_{\text{eff}}, \Gamma_{\text{decay}}, \tau_{\text{life}}, \Theta, \text{Cert}_{\text{vac}}, (\rho_{\text{vac}}^{\text{raw}}, \rho_{\text{vac}}^{\text{route}}, \rho_{\text{vac}}^{\text{stand}})),$$

where V_{eff} is an effective-potential presentation, Γ_{decay} decay or tunneling data, τ_{life} the lifetime comparison, Θ the thermal or route environment, Cert_{vac} the relevant gauge, scheme, matching, lifetime, thermal, and comparison certificate, and $(\rho_{\text{vac}}^{\text{raw}}, \rho_{\text{vac}}^{\text{route}}, \rho_{\text{vac}}^{\text{stand}})$ the layered raw-role, route-status, and standing-status classifiers.

The main theorem is the Higgs Vacuum Endpoint Theorem:

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Vac}_H) = 0, \quad \text{Surv}_{D_H}^{\text{prim}}(\text{LowerBranch}_H) = 0.$$

Route availability does not imply standing closure:

$$\text{RouteAvail}_{D_H}(v) \not\Rightarrow \text{StandingClosed}_{D_H}(v).$$

A route may be available, calculable, and physically meaningful without becoming a primitive Higgs ground and without reviving primitive lower-branch standing. Standard metastability, effective-potential, thermal, and electroweak-baryogenesis discussions are therefore preserved as route or certificate questions, while primitive vacuum standing is deleted. Richer-scope mechanisms remain admissible only as declared target updates with their own carrier, descent, and witness burdens.

Keywords: AASC; Higgs vacuum; route status; metastability; false vacuum; lower branch; route availability; standing closure; effective potential; lifetime normal form; electroweak crossover; electroweak baryogenesis.

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Mathematical-status lock and proof-class boundary

This manuscript is an AASC endpoint-classification theorem for Higgs-vacuum claims inside the Higgs structural endpoint arc. It is not a new Standard Model vacuum-stability calculation, not a new bounce-action computation, not a new finite-temperature lattice result, not a new electroweak-baryogenesis calculation, not a claim that metastability is false, and not an anti-BSM theorem. Its contribution is the status calculus that separates vacuum description, route availability, standing closure, primitive promotion, and target update after C_H , Higgs identity, and Higgs scale standing have already been fixed.

The proof class is downstream. Paper 1 supplies D_H , C_H , H_{role} , representative collapse, and the local Higgs standing filter. Paper 2 supplies the identity endpoint $\text{Id}_H = \text{Occ}_{D_H}(C_H)$. Paper 3 supplies the no-primitive-scale result and carrier-indexed scale normal form R_H^{scale} . This paper evaluates only vacuum and route claims on top of those packages. A same-target objection must break one of the following: the inherited carrier package, the identity endpoint, the scale-standing dependency, the vacuum claim taxonomy, the route-normal-form construction, the route-availability/non-closure distinction, the lower-branch exhaustion, the thermal/EWBG route classification, the richer-scope boundary, or the final primitive-survivor inference.

Read this first: endpoint proof spine

The proof has seven layers.

Layer I: inherited carrier, identity, and scale. Higgs identity is already fixed as $\text{Occ}_{D_H}(C_H)$, and Higgs scale claims are already carrier-indexed. A vacuum branch cannot define the carrier or the scale by which it is evaluated.

Layer II: vacuum claim taxonomy. Submitted vacuum claims are separated into present-vacuum descriptions, potential minima, lower branches, metastability, instability crossings, lifetime routes, tunneling/bounce routes, thermal routes, first-order electroweak-transition routes, EWBG-style routes, and richer-scope mechanisms.

Layer III: representative collapse. Vacuum location, potential shape, branch membership, field-coordinate location, and thermal profile are representatives or route data. They are not primitive Higgs identity grounds or primitive Higgs scale grounds.

Layer IV: route normal form. A same-scope standing-bearing vacuum claim must factor through R_H^{vac} . The lifetime specialization is R_{life} , which contains carrier, scale transport, effective-potential presentation, decay/lifetime data, comparison class, and certificate.

Layer V: route availability versus standing closure. RouteAvail records that a route is well-typed and evaluable in the declared comparison class. StandingClosed records that the route plus certificate closes a standing-bearing endpoint status. The implication from availability to closure is invalid.

Layer VI: primitive lower-branch deletion. A lower branch may be diagnostic, route-available, lifetime-relevant, or target-updating. It is never a primitive same-scope Higgs ground. Thus $\text{Surv}_{D_H}^{\text{prim}}(\text{LowerBranch}_H) = 0$.

Layer VII: thermal and richer-scope classification. Thermal routes and EWBG-style routes are route statuses or target updates. They do not restore primitive branch standing. A successful BSM or richer-scope route must declare its carrier inventory, target update, descent map, and witness package.

1 Introduction: vacuum is not identity

Vacuum language is powerful because it looks foundational. It speaks of minima, branches, tunneling, false-vacuum decay, thermal history, bubble nucleation, and electroweak-baryogenesis routes. But after Papers 1–3, those objects cannot answer the carrier or identity question. The Higgs is already fixed structurally by C_H . A vacuum claim can classify a downstream route associated with C_H ; it cannot define C_H or become an independent primitive Higgs ground.

The central distinction is:

$$\text{route availability} \neq \text{standing closure}.$$

A route can be well-typed, calculable, and significant while falling short of a standing endpoint. A metastability calculation can be an admissible lifetime route without converting a lower branch into primitive standing. A thermal route can classify a finite-temperature transition without redefining Higgs identity. A first-order or EWBG route can be a richer-scope update without becoming a same-scope survivor of the minimal carrier target.

This paper therefore preserves the physics language while controlling its status. It does not deny effective potentials, RG-improved vacuum analyses, bounce calculations, finite-temperature transition studies, or electroweak-baryogenesis route conditions. It denies only primitive promotion of their representative components.

The sentence that governs the manuscript is:

A vacuum route may be available, calculable, and physically meaningful without becoming a primitive Higgs ground.

2 Dependency paragraph and inherited package

2.1 Inherited carrier and identity

This paper imports the local Higgs package established by Papers 1–3:

$$D_H, \quad C_H, \quad H_{\text{role}}, \quad R_H, \quad \text{SF}_H,$$

with Higgs identity fixed by

$$\text{Id}_H = \text{Occ}_{D_H}(C_H).$$

It also imports the scale result:

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Scale}_H) = 0,$$

and the scale normal form R_H^{scale} . Running parameters, instability scales, and effective-potential field coordinates may appear in vacuum routes only as carrier-indexed scale or route data.

2.2 Local task of Paper 4

The local task is not to re-evaluate the Standard Model vacuum numerically. The local task is to classify the standing status of vacuum claims. The paper asks:

Given C_H , Id_H , and R_H^{scale} , what can a Higgs-vacuum claim be?

The answer is a route-status theorem. Same-scope Higgs-vacuum claims are representatives, route markers, route-available protocols, standing-closed certificate statuses, standing-smuggling failures, target updates, or failed primitives. No primitive vacuum ground remains.

2.3 Dependency use table

Table 1: Imported dependencies and local use in Paper 4

Imported item	Source role	Work performed here
C_H	Higgs bridge/fixation carrier	Prevents vacuum descriptions from defining Higgs identity.
$\text{Id}_H = \text{Occ}(C_H)$	Structural identity endpoint	Makes vacuum status downstream rather than identity-grounding.
R_H^{scale}	Carrier-indexed scale normal form	Forces running, crossing, and instability scales to enter as route data.
Fixed-domain exhaustion	No further same-scope governance locus	Supports route-status and lower-branch exhaustion.
No-generator/no-transfer/no-repair	Standing conservation discipline	Blocks route success from becoming standing closure by transfer.
No primitive lower-vacuum endpoint support	Vacuum endpoint precedent	Supplies a detailed companion route analysis for measured minimal-SM lower-branch claims.

3 Vacuum claim taxonomy

Definition 3.1 (Higgs-vacuum claim). A Higgs-vacuum claim is a submitted claim $\text{VacClaim}_H(v)$ asserting that v fixes, classifies, explains, routes, or certifies a Higgs-vacuum status associated with the fixed Higgs carrier C_H in the same-scope Higgs-domain target D_H .

Definition 3.2 (Primitive vacuum claim). A primitive vacuum claim is a submitted claim that purports to carry same-scope Higgs-vacuum standing directly as a vacuum ground, rather than as a representative, route marker, route-available protocol, certificate-closed downstream status, or declared target update.

Definition 3.3 (Vacuum claim catalogue). The local catalogue contains the following submitted claim forms:

1. present-vacuum description;
2. potential minimum or effective-potential profile;
3. lower-branch membership;
4. metastability or instability status;
5. instability crossing or running-scale marker;
6. lifetime, tunneling, or bounce route;

7. finite-temperature transition route;
8. first-order electroweak phase-transition route;
9. electroweak-baryogenesis-style route;
10. richer-scope vacuum mechanism.

The catalogue is not an empirical survey. It is the fixed-domain list of functions by which a vacuum claim can alter or attempt to alter standing status.

3.1 Taxonomy table

Table 2: Vacuum claim taxonomy and route-status classification

Claim form	What it can legitimately do	Primitive failure mode	Allowed status
Present vacuum description	Name the carrier-associated electroweak vacuum in a presentation.	Does not fix C_H or identity by itself.	Representative.
Potential minimum	Present a scalar/effective-potential chart.	Gauge, scheme, matching, and coordinate dependence unless quotiented.	Representative or diagnostic.
Lower branch	Supply a candidate lower channel.	Branch membership is not endpoint standing.	Route marker or lower-branch nonprimitive.
Metastability	Classify lifetime status.	Requires lifetime comparison and certificate.	Route-available or standing-closed under certificate.
Instability crossing	Mark running or effective-potential transition.	Paper 3 deletes primitive scale standing.	Route marker.
Bounce/lifetime route	Evaluate decay probability or lifetime.	Availability does not close standing.	Route-available, route-incomplete, or standing-closed certificate.
Thermal route	Classify finite-temperature behavior.	Thermal history is downstream, not identity.	Thermal route status.
First-order EWPT route	Assert barrier/transition structure.	Not same-scope if it requires new carriers.	Route status or target update.
EWBG route	Assert production/preservation route.	Needs departure, CP/source, transport, and washout package.	Route status or target update.

Claim form	What it can legitimately do	Primitive failure mode	Allowed status
Richer-scope mechanism	Add scalar, threshold, UV, gravitational, or source content.	Leaves same-scope target.	Target update.

Table 3: Vacuum functions and fixed-domain loci

Vacuum function	Fixed-domain locus	Classifier layer	Status result
Describe a vacuum	Lawful quotient / representative skin	Raw role	Vacuum representative.
Mark a branch or crossing	Admissible continuation marker	Raw role and route marker	Branch marker, Running marker, or Route marker.
Transport/run data	Carrier-indexed scale normal form	Route status	Route marker or Route available only if a protocol is declared.
Evaluate lifetime	Lifetime normal form	Route status	Route available, Route incomplete, or Route closed under certificate.
Close standing	Certificate partition / standing quotient	Standing status	Carrier-indexed certified vacuum content.
Select lower branch as primitive	Standing partition attempt	Standing status	Lower-branch nonprimitive or Standing smuggling.
Classify thermal/EWBG route	Typed finite-temperature or source route environment	Raw role and standing status	Thermal route or Target update.
Change physics or carrier inventory	Boundary typing / invariant bundle	Standing status	Target update.
Assert primitive vacuum ground	No standing locus after route exhaustion	Standing status	Failed primitive.

Lemma 3.4 (Vacuum taxonomy completeness). *Every submitted same-scope Higgs-vacuum claim in the Paper-4 target receives a layered classification: a raw descriptive role where applicable, a route status where a protocol or marker is asserted, and a standing status for same-scope closure, failure, or target update. Descriptive role, route availability, and standing closure are not rival statuses.*

Proof. A vacuum claim can perform standing-relevant work only by changing one of the fixed-domain loci listed in the preceding table. Description changes the representative layer. Marking, transport, and lifetime evaluation change the route layer. Certificate closure changes the standing partition. Primitive selection attempts change the standing partition without certificate and are classified as lower-branch nonprimitivity, standing-smuggling, or failed primitive. Thermal and EWBG submissions enter a typed route environment, and carrier/source changes are target updates. A further same-scope function would have to alter vacuum-standing work while changing none of description, route, certificate, boundary typing, or primitive-assertion loci; fixed-domain exhaustion forbids such a residue. $\square$

4 Vacuum representative collapse

Definition 4.1 (Vacuum representative). A vacuum representative is a coordinate, chart, potential profile, field-space picture, branch label, thermal profile, or diagnostic object used to present a vacuum-related calculation while preserving the inherited carrier C_H . It carries no primitive standing merely by being writable or useful.

Lemma 4.2 (Vacuum-description separation). *For any submitted vacuum description v ,*

$$\text{VacRepresentative}(v) \not\Rightarrow \text{Stand}_{D_H}^{\text{prim}}(v).$$

Proof. A vacuum representative can display branch structure, potential shape, local minima, thermal profiles, or running-scale markers. None of those functions supplies the inherited carrier, the identity endpoint, or a standing verdict. If the representative is used to support a stability or lifetime claim, it must enter a route normal form. If it is used to support a thermal claim, it must enter the thermal route. If it requires added carrier or boundary content, it exits to a target update. No representative description, as such, supplies primitive vacuum standing. $\square$

Theorem 4.3 (Vacuum representative collapse). *Vacuum location, potential shape, effective-potential branch membership, field-coordinate location, instability crossing, thermal profile, and lower-branch depiction do not supply primitive Higgs identity, primitive Higgs scale standing, or primitive Higgs-vacuum standing.*

Proof. Higgs identity is already fixed by $\text{Occ}_{D_H}(C_H)$. Paper 3 deletes primitive Higgs-scale standing and requires scale representatives to be carrier-indexed. A vacuum representative can therefore contribute only downstream route data. Potential shape and field-coordinate location are representative data. Branch membership is a route marker until a static, lifetime, or thermal verdict is certified. Instability crossings are running-scale route markers. Thermal profiles belong to a finite-temperature route. If any such object is promoted directly to primitive standing, it bypasses the carrier, identity, scale, or route certificate. That is standing-smuggling. Therefore the representative classes collapse as primitive grounds. $\square$

5 Vacuum route normal form

Definition 5.1 (Vacuum route normal form). The vacuum route normal form is

$$R_H^{\text{vac}} = (D_H, C_H, R_H^{\text{scale}}, V_{\text{eff}}, \Gamma_{\text{decay}}, \tau_{\text{life}}, \Theta, \text{Cert}_{\text{vac}}, \rho_{\text{vac}}).$$

Here D_H is the inherited Higgs-domain target, C_H the Higgs carrier, R_H^{scale} the scale-normal-form package used for running and crossing data, V_{eff} an effective-potential or free-energy presentation, Γ_{decay} decay/tunneling data when present, τ_{life} the lifetime comparison when present, Θ the thermal or route environment, Cert_{vac} the gauge/scheme/matching/lifetime/thermal certificate demanded by the asserted route, and $(\rho_{\text{vac}}^{\text{raw}}, \rho_{\text{vac}}^{\text{route}}, \rho_{\text{vac}}^{\text{stand}})$ the layered raw-role, route-status, and standing-status classifiers.

Definition 5.2 (Vacuum route normalizers). The route normalizer $\text{NF}_{H,\text{vac}}^{\text{route}}(v)$ is defined when the submitted claim preserves D_H and C_H , uses scale material only through R_H^{scale} , declares the route environment, and provides the route data needed to make the protocol evaluable. When defined, $\text{NF}_{H,\text{vac}}^{\text{route}}(v)$ records the route fields of R_H^{vac} instantiated by the submitted protocol.

The standing normalizer $\text{NF}_{H,\text{vac}}^{\text{stand}}(v)$ is defined only when $\text{NF}_{H,\text{vac}}^{\text{route}}(v) \downarrow$, the relevant closure certificate Cert_{vac} is supplied, and the submission does not promote route grammar into primitive standing. When defined, $\text{NF}_{H,\text{vac}}^{\text{stand}}(v) = R_H^{\text{vac}}$ with both route and certificate fields instantiated.

Theorem 5.3 (Vacuum normal-form separation). *For any same-scope Higgs-vacuum claim v ,*

$$\text{RouteAvail}_{D_H}(v) = 1 \Rightarrow \text{NF}_{H,\text{vac}}^{\text{route}}(v) \downarrow,$$

and

$$\text{StandingClosed}_{D_H}(v) = 1 \Rightarrow \text{NF}_{H,\text{vac}}^{\text{stand}}(v) \downarrow \wedge \text{RouteAvail}_{D_H}(v) = 1.$$

Moreover, $\text{NF}_{H,\text{vac}}^{\text{route}}(v) \downarrow$ does not imply $\text{NF}_{H,\text{vac}}^{\text{stand}}(v) \downarrow$.

Proof. Route availability requires carrier preservation, carrier-indexed scale use, a declared route environment, and evaluable route data. These are precisely the route fields recorded by $\text{NF}_{H,\text{vac}}^{\text{route}}$. Standing closure adds the certificate field: gauge, scheme, matching, comparison, uncertainty, witness, and route-specific closure data. Therefore closure implies both route availability and $\text{NF}_{H,\text{vac}}^{\text{stand}}$. The converse fails because an evaluable route may lack the required certificate components. Omitting the certificate leaves route availability without standing closure rather than a weaker form of standing. $\square$

Theorem 5.4 (Vacuum standing normal form theorem). *If a same-scope Higgs-vacuum claim is standing-bearing, then its vacuum content factors through R_H^{vac} :*

$$\text{StandingClosed}_{D_H}(v) \Rightarrow \text{NF}_{H,\text{vac}}^{\text{stand}}(v) \downarrow.$$

Proof. Standing closure requires the claim to preserve the inherited carrier and identity target, to use running or scale data only through carrier-indexed transport, to identify whether the route is static, lifetime, thermal, or richer-scope, and to supply a certificate appropriate to that route. Those are precisely the standing fields of R_H^{vac} . Omitting the carrier destroys same-scope Higgs identity. Omitting scale-normal form revives primitive scale standing. Omitting route data leaves only description. Omitting the certificate leaves availability without closure. Hence every standing-bearing vacuum claim factors through the standing normal form. $\square$

6 Route availability versus standing closure

Definition 6.1 (Route availability). $\text{RouteAvail}_{D_H}(v) = 1$ means that the submitted vacuum claim has a well-typed route in the declared target: the carrier is preserved, representative and scale data are admissible, the route environment is declared, and the protocol is evaluable as a candidate route. Equivalently, $\text{NF}_{H,\text{vac}}^{\text{route}}(v) \downarrow$.

Definition 6.2 (Standing closure). $\text{StandingClosed}_{D_H}(v) = 1$ means that the submitted route has not only availability but also the certificate needed to produce a standing-bearing downstream status in the fixed target. Equivalently, $\text{NF}_{H,\text{vac}}^{\text{stand}}(v) \downarrow$ and the asserted standing verdict is the certificate-mediated verdict of the route.

Lemma 6.3 (Availability without closure witness class). *There are same-scope Higgs-vacuum route submissions that are route-available but not standing-closed.*

Proof. A lifetime submission may declare C_H , R_H^{scale} , V_{eff} , Γ_{decay} , and τ_{life} , while lacking the uncertainty, comparison, or rival-deletion fields of $\text{Cert}_{\text{life}}$. Such a submission is evaluable as a route but not standing-closed. A lower-branch submission may supply RG or effective-potential branch data while lacking the singleton/static/lifetime certificate needed to select an endpoint. A thermal or EWBG-style submission may declare a finite-temperature or source route while lacking the target-preserving endpoint, source, transport, or washout certificate. Each case defines $\text{NF}_{H,\text{vac}}^{\text{route}}$ but not $\text{NF}_{H,\text{vac}}^{\text{stand}}$. $\square$

Theorem 6.4 (Route availability non-closure). *Route availability does not imply standing closure:*

$$\text{RouteAvail}_{D_H}(v) = 1 \not\Rightarrow \text{StandingClosed}_{D_H}(v) = 1.$$

Proof. Route availability is a form-of-route status. It records that a protocol can be entered and evaluated within the declared target. Standing closure is a verdict status. It requires the relevant certificate: for lifetime routes, a lifetime comparison and uncertainty certificate; for thermal routes, a thermodynamic route certificate; for EWBG-style routes, source/transport/washout closure; and for richer-scope routes, a target-update and descent certificate. The witness classes in the preceding lemma show route availability without closure. Treating the route as closed merely because it is available transfers standing from route grammar to endpoint verdict. The no-generator/no-transfer/no-repair discipline blocks that transfer. Therefore the implication is invalid. $\square$

Corollary 6.5 (Route availability is compatible with non-selection). *A vacuum route can be available, calculable, and physically meaningful while the corresponding endpoint is not selected.*

Proof. Selection requires certified deletion of inequivalent rivals or a declared downstream certificate. Availability supplies neither rival deletion nor certificate closure. Therefore availability and non-selection are compatible. $\square$

Corollary 6.6 (No route laundering). *A lower branch, thermal profile, or EWBG-style protocol cannot launder route availability into primitive Higgs identity, primitive scale, or primitive vacuum standing.*

Proof. Identity and scale standing are already fixed upstream, and primitive vacuum standing is the local target of deletion. A route can produce only the status certified by its normal form. It cannot transfer standing backward into the carrier, identity, or primitive scale layer. $\square$

7 Lifetime normal form

Definition 7.1 (Lifetime route normal form). The lifetime route normal form is

$$R_{\text{life}} = (D_H, C_H, R_H^{\text{scale}}, V_{\text{eff}}, T_{\text{RG}}, \Gamma_{\text{decay}}, \tau_{\text{life}}, \mathcal{C}_{\text{life}}, \text{Cert}_{\text{life}}, \rho_{\text{route}}),$$

where T_{RG} is the running, scheme, matching, and transport package; Γ_{decay} includes bounce or decay-rate data; τ_{life} declares the comparison time or spacetime-volume criterion; $\mathcal{C}_{\text{life}}$ is the lifetime comparison class; $\text{Cert}_{\text{life}}$ contains gauge, scheme, top-mass, matching, perturbative, continuation, uncertainty, and comparison-envelope components; and ρ_{route} is the route classifier.

Definition 7.2 (Lifetime route classifier). A lifetime route receives one of six local statuses: route-normal, route-incomplete, route marker, standing-smuggling, target update, or failed primitive. A route-normal protocol supplies the components of R_{life} . A route-incomplete protocol preserves target but lacks at least one component. A route marker supplies diagnostic or branch data without a route. Standing-smuggling treats availability as closure. A target update changes the domain. A failed primitive assertion attempts to use vacuum status directly as a Higgs ground.

Lemma 7.3 (Forced lifetime components). *A same-scope lifetime claim capable of standing closure requires every field of R_{life} .*

Proof. The target and carrier fields preserve Higgs identity. R_H^{scale} and T_{RG} prevent running and crossing data from becoming primitive scale grounds. V_{eff} supplies the branch or potential presentation. Γ_{decay} and τ_{life} supply the decay and comparison data. The comparison class specifies what lifetime threshold is being asserted. $\text{Cert}_{\text{life}}$ provides the gauge, scheme, matching, perturbative, continuation, and uncertainty controls. Without any one component, the lifetime claim is either not same-scope, not evaluable, or not standing-closed. $\square$

Theorem 7.4 (Lifetime normal form theorem). *For any same-scope lifetime claim ℓ ,*

$$\text{StandingClosed}_{D_H}(\ell) = 1 \Rightarrow \text{NF}_{\text{life}}(\ell) \downarrow \wedge \ell \cong_{D_H} R_{\text{life}}.$$

Proof. If the lifetime claim is standing-closed, it must preserve D_H , C_H , and carrier-indexed scale transport. It must declare the potential representation, decay/tunneling data, lifetime comparison, and certificate. Those fields instantiate R_{life} . Up to notation, scheme presentation, or representative bookkeeping that does not alter route status, the claim is equivalent to its lifetime normal form. $\square$

8 Lower-branch standing exhaustion

Definition 8.1 (Lower-branch claim). A lower-branch claim $\text{LowerBranch}_H(b)$ is a submitted claim that a lower effective-potential branch, lower vacuum channel, or lower field-space region supplies Higgs-vacuum standing.

Definition 8.2 (Lower-branch functions). A lower branch can perform exactly four admissible route functions inside this paper’s target:

1. static diagnostic: it displays a candidate lower channel;
2. lifetime input: it supplies a possible decay channel for R_{life} ;
3. thermal input: it contributes to a finite-temperature route;
4. target update: it becomes standing-bearing only after new carrier, source, or boundary content is declared.

These are route functions, not primitive standing grounds.

Lemma 8.3 (Lower branch does not fix carrier, identity, or scale). *For any lower-branch representative b ,*

$$\text{LowerBranch}_H(b) \not\Rightarrow C_H, \quad \text{LowerBranch}_H(b) \not\Rightarrow \text{Id}_H, \quad \text{LowerBranch}_H(b) \not\Rightarrow \text{Stand}^{\text{prim}}(\text{Scale}_H).$$

Proof. The carrier and identity are inherited from Papers 1 and 2. The scale-standing result is inherited from Paper 3. A lower branch may use carrier-indexed scale data or supply route information, but it does not generate those upstream statuses. Thus the displayed branch cannot fix C_H , Id_H , or primitive scale standing. $\square$

Theorem 8.4 (No primitive lower branch). *The lower branch has no same-scope primitive survivor:*

$$\text{Surv}_{D_H}^{\text{prim}}(\text{LowerBranch}_H) = 0.$$

Proof. Let b be a submitted lower-branch claim. If b supplies only a potential or field-coordinate picture, it is a representative or route marker. If it supplies a lifetime route, it must factor through R_{life} , and the standing-bearing object is the certified lifetime status, not the branch itself. If it supplies a thermal route, the standing-bearing object is a thermal route status, not primitive branch standing. If it requires new carrier, threshold, scalar, gravitational, or source content, it is a target update. If it attempts to bypass these routes, it is standing-smuggling or failed primitive. The fixed-domain exhaustion theorem blocks a further same-scope lower-branch function. Hence no primitive lower-branch survivor remains. $\square$

Corollary 8.5 (Metastability does not revive lower-branch primitivity). *A certified metastability verdict, if supplied, selects or classifies C_{meta} as a downstream lifetime status. It does not select C_{lower} as primitive standing.*

Proof. Metastability is a lifetime verdict under a comparison criterion. Its standing, when certified, is mediated by R_{life} . A mediated verdict cannot be retyped as primitive branch standing without reversing the dependency order. $\square$

9 Instability crossings and running-scale claims

Definition 9.1 (Instability route marker). An instability route marker is a running-coupling crossing, effective-potential crossing, or field-coordinate indicator, such as $\lambda(\mu) = 0$, Λ_{inst} , or $V_{\text{eff}}(\phi)$ developing a lower region, used as diagnostic input to a vacuum route.

Theorem 9.2 (Instability crossing non-primitive). *Instability crossings and running-scale markers are not primitive Higgs-vacuum endpoints.*

Proof. Paper 3 has already shown that running and crossing scales cannot carry primitive Higgs-scale standing. In a vacuum route, those quantities can function only as carrier-indexed transport data or route markers. If a crossing is used for a static, lifetime, or thermal claim, the claim must enter R_H^{vac} . If it is used directly as endpoint standing, it bypasses carrier-indexed scale status and route certification. That is standing-smuggling. Therefore instability crossings are nonprimitive. $\square$

Corollary 9.3 (No scale-to-vacuum shortcut).

$$\Lambda_{\text{inst}} \not\Rightarrow \text{StandingClosed}_{D_H}(\text{Vac}_H).$$

Proof. A scale marker does not contain a lifetime criterion, decay probability, thermal verdict, rival deletion, or richer-scope certificate. It therefore cannot close vacuum standing by itself. $\square$

10 Thermal and EWBG route classification

Definition 10.1 (Thermal route). A thermal route is a submitted vacuum claim whose standing-relevant content depends on finite-temperature effective potentials, free energies, dimensional-reduction or lattice data, transition order, bubble-wall dynamics, sphaleron/washout structure, or finite-temperature source terms.

Definition 10.2 (EWBG-style route). An EWBG-style route is a submitted route claiming that electroweak-sector dynamics produce and preserve a baryon asymmetry through some combination of baryon-number violation, CP/source structure, departure from equilibrium, transport, and washout avoidance.

Theorem 10.3 (Thermal route classification). *A thermal route is a route status or target update; it is not a primitive Higgs-vacuum ground:*

$$\text{ThermalRoute}(t) \not\Rightarrow \text{Stand}_{D_H}^{\text{prim}}(\text{Vac}_H).$$

Proof. A thermal route belongs to a finite-temperature comparison environment Θ . If it preserves the same target and supplies a route protocol, it is a thermal route status. If it supplies sufficient thermodynamic certificate, it may close a downstream thermal status. If it introduces new scalar, threshold, source, or thermal-boundary content, it changes target. None of these alternatives makes the thermal profile a primitive Higgs ground. $\square$

Theorem 10.4 (EWBG route classification). *An EWBG-style route is not a primitive Higgs-vacuum route. It is a production-and-preservation route with its own source, transport, departure, and washout burdens.*

Proof. An EWBG-style route asks whether an observed-asymmetry production and preservation protocol is available and certified. That target is distinct from Higgs identity, Higgs scale standing, and lower-branch primitive standing. If the route is same-scope, it must supply its route conditions. If it succeeds only by adding CP sources, scalar sectors, threshold structure, or thermal dynamics outside the target, it is a target update. In neither case does the route become primitive vacuum standing. $\square$

Corollary 10.5 (No thermal laundering of lower-branch standing). *A finite-temperature route cannot convert a zero-temperature lower branch into primitive same-scope Higgs-vacuum standing.*

Proof. The route target has changed from lower-branch depiction to thermal route status. A route-status change cannot retroactively convert the original branch representative into primitive standing. If new carrier content is added, the claim is a target update rather than same-scope revival. $\square$

11 Richer-scope escape conditions

Definition 11.1 (Richer-scope vacuum update). A richer-scope vacuum update is a declared move from D_H^{vac} to $D_{H,+}^{\text{vac}}$ by adding or modifying at least one standing-relevant item: carrier inventory, scalar sector, threshold structure, CP/source ledger, thermal target, UV boundary condition, gravitational boundary condition, cosmological source condition, comparison class, or admissibility-relevant invariant bundle.

Lemma 11.2 (Richer-scope boundary). *If a vacuum claim requires new carrier, threshold, scalar, thermal, source, UV, gravitational, or cosmological content in order to close standing, then it is not a same-scope counterexample to primitive vacuum deletion. It is a target update.*

Proof. Same-scope evaluation preserves the inherited carrier, identity, scale, and admissibility boundary. A new carrier, threshold, scalar sector, source ledger, thermal target, UV boundary, or gravitational rule changes the invariant bundle under which the route is evaluated. Such a claim may be admissible in a richer target, but it does not revive same-scope primitive standing. $\square$

Theorem 11.3 (Richer-scope escape normal form). *A successful richer-scope vacuum route must supply*

$$(\Sigma_{\text{new}}, D_H^{\text{vac}} \rightarrow D_{H,+}^{\text{vac}}, C_H^+, (R_H^{\text{scale}})^+, (R_H^{\text{vac}})^+, \text{Cert}^+, \text{Witness}^+),$$

where Σ_{new} is the new carrier/source content, C_H^+ the updated carrier relation, $(R_H^{\text{scale}})^+$ and $(R_H^{\text{vac}})^+$ updated scale and vacuum normal forms, Cert^+ the closure certificate, and Witness^+ the empirical, structural, or comparison witness package.

Proof. A richer-scope claim is not merely a more detailed representative. It changes the target. To remain admissible, it must declare what changed, how the Higgs carrier relation is preserved or replaced, how scale and vacuum routes are transported, what certificate closes standing, and what witnesses ground the update. Those are exactly the fields listed. $\square$

12 Vacuum status classifier

Definition 12.1 (Layered vacuum classifier). The vacuum classifier has three layers.

1. The raw-role classifier $\rho_{\text{vac}}^{\text{raw}}(v)$ records descriptive function: Vacuum representative, Branch marker, Running marker, Lifetime protocol, Thermal protocol, EWBG protocol, or Target-update submission.
2. The route-status classifier $\rho_{\text{vac}}^{\text{route}}(v)$, where applicable, records No route, Route marker, Route incomplete, Route available, or Route closed under certificate.
3. The standing-status classifier $\rho_{\text{vac}}^{\text{stand}}(v)$ records Carrier-indexed certified vacuum content, Lower-branch nonprimitive, Thermal route only, Standing smuggling, Target update, or Failed primitive.

Representatives, markers, and protocols are descriptive or route-layer statuses. Route availability is route-form status. Carrier-indexed certified vacuum content is the successful standing object, and certificate closure is the mode by which that object acquires same-scope standing. Standing-smuggling, target-update, and failed-primitive statuses are failure, update, or deletion statuses, not standing closures. A single submission may have a raw role and a route status while receiving only one standing-status classification.

Theorem 12.2 (Layered vacuum classifier totality). *Every submitted Higgs-vacuum claim receives one raw role, one route status where a route function is asserted, and one standing-status classification. Descriptive role, route availability, and standing closure are not rival statuses.*

Proof. By the taxonomy completeness lemma and the vacuum-function table, every claim changes a raw descriptive function, a route function, a standing/certificate function, a boundary/update function, or no standing locus. Raw descriptive functions are assigned by $\rho_{\text{vac}}^{\text{raw}}$. Route functions are assigned by $\rho_{\text{vac}}^{\text{route}}$. Standing closure, failure, lower-branch deletion, standing-smuggling, and target updates are assigned by $\rho_{\text{vac}}^{\text{stand}}$. Because the layers answer different questions—what the object describes, whether a route is evaluable, and what standing status follows—overlap between a representative and an available route does not make the classifier nonfunctional. A certified lifetime route can be both a Lifetime protocol at the raw layer and Route closed under certificate at the route layer while receiving Carrier-indexed certified vacuum content at the standing layer. No further same-scope route function remains because any such function must alter standing without changing any fixed-domain locus already recorded by the three layers. $\square$

Table 4: Layered vacuum statuses and fixed-domain loci

Status or layer	Layer	Fixed-domain locus	Result
Vacuum representative / Branch marker / Running marker	raw role	lawful quotient / skin	Descriptive or marker content only.
Lifetime protocol / Thermal protocol / EWBG protocol	raw role	declared route environment	Protocol submission, not closure.
Route marker / Route incomplete / Route available	route	admissible continuation / route normal form	Evaluable or incomplete route status.
Route closed under certificate	route	certificate partition	Route has supplied closure fields.
Carrier-indexed certified vacuum content	standing	certified quotient	Same-scope downstream standing; the surviving object is carrier-indexed and the closure mode is certificate-mediated.
Lower-branch nonprimitive / Thermal route only	standing	no-promotion / typed route split	Route status without primitive standing.
Standing smuggling / Failed primitive	standing	standing partition / no standing locus	Failed primitive promotion.
Target update	standing	boundary typing	Claim exits the fixed target.

13 Vacuum endpoint theorem

Theorem 13.1 (Higgs Vacuum Endpoint Theorem). *Every submitted same-scope Higgs-vacuum claim downstream of C_H , ld_H , and R_H^{scale} receives the layered classification of Definition 12.1. Among same-scope standing-bearing vacuum claims, only carrier-indexed, certificate-mediated vacuum content survives. No primitive vacuum ground survives:*

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Vac}_H) = 0.$$

More specifically,

$$\text{Surv}_{D_H}^{\text{prim}}(\text{LowerBranch}_H) = 0.$$

Proof. Let v be a submitted vacuum claim. The layered classifier assigns the raw role, route status, and standing status of v . If v is only a representative or route marker, it is not primitive. If it is route-available, route availability does not imply standing closure. If it is standing-closed, its standing is certificate-mediated and carrier-indexed, not primitive. If it asserts lower-branch standing, the no-primitive-lower-branch theorem deletes it. If it is thermal or EWBG-style, it is a typed route status or a target update. If it smuggles standing, it is rejected by no-generator/no-transfer/no-repair discipline. If it changes carrier or comparison boundary, it is a target update. If none of those statuses applies, it has no standing-bearing function. Thus no primitive vacuum ground survives. The lower branch is a special case already exhausted by the lower-branch theorem. $\square$

Corollary 13.2 (Standard vacuum calculations are preserved). *The theorem does not reject vacuum-stability calculations, metastability analyses, effective-potential diagnostics, or finite-temperature route studies. It classifies their standing route.*

Proof. The theorem admits representatives, markers, route availability, and standing closure under certificate. It deletes only primitive promotion. $\square$

Corollary 13.3 (No primitive vacuum recovery through future precision). *Future precision may close a certificate or update a source state. It cannot retroactively convert an earlier route marker or lower branch into primitive vacuum standing in the same act.*

Proof. Future closure is a new certificate or source-state update. Irreversibility blocks same-act retroactive repair. $\square$

14 Relation to standard physics language

The manuscript preserves ordinary vocabulary by typing it.

- **Effective potential.** Admitted as a representative and diagnostic; endpoint standing requires quotient or route descent.
- **Lower branch.** Admitted as a branch diagnostic or route marker; primitive standing is deleted.
- **Metastability.** Admitted as lifetime-route status; standing closure requires R_{life} and $\text{Cert}_{\text{life}}$.
- **Instability scale.** Admitted as carrier-indexed running/route marker; primitive scale and primitive endpoint readings are deleted.
- **Thermal transition.** Admitted as finite-temperature route status; not a primitive identity or branch ground.
- **EWBG route.** Admitted as a route target with source, transport, departure, and washout burdens; not a primitive vacuum route.
- **BSM route.** Admitted only by target update with carrier, descent, and witness package.

The result therefore does not compete with numerical physics. It classifies raw vacuum roles, route statuses, standing-closed certificates, and target updates without treating descriptive or failure classes as forms of standing.

15 Handoff to extension classifier

Paper 5 classifies extra scalar, composite, role-splitting, and BSM Higgs-domain claims. Paper 4 supplies one input to that classifier: vacuum mechanisms that require new carrier inventory or source structure are not same-scope counterexamples. They are extension candidates. They must be evaluated by carrier preservation, role occupation, scale transport, vacuum-route descent, and witness closure. This handoff is not a research plan; it is the formal boundary of the current theorem.

16 Conclusion

The Higgs structural arc now has four local deletions. Paper 1 deletes primitive representative carrier standing. Paper 2 deletes primitive representative identity standing. Paper 3 deletes primitive Higgs scale standing. Paper 4 deletes primitive Higgs-vacuum standing. Vacuum descriptions, lower branches,

lifetime routes, thermal profiles, and EWBG-style routes remain important, but their standing is downstream and typed.

The closed endpoint of this paper is:

$$\text{Surv}_{D_H}^{\text{prim}}(\text{Vac}_H) = 0, \quad \text{Surv}_{D_H}^{\text{prim}}(\text{LowerBranch}_H) = 0, \quad \text{RouteAvail}_{D_H}(v) \not\Rightarrow \text{StandingClosed}_{D_H}(v).$$

A vacuum representative may describe; a route may be available; only certificate-closed carrier-indexed content bears same-scope standing. A lower branch may be diagnostic without being primitive. A thermal route may classify history without defining the Higgs. A richer-scope mechanism may be admissible only after it declares a new target and closure certificate. This is the vacuum endpoint of the Higgs-domain structural program.

A Vacuum route taxonomy register

Table 5: Vacuum route taxonomy register

Object	Route role	Required certificate for closure	Primitive verdict
Present vacuum	Reference representative	Carrier and identity already inherited	Not primitive.
Potential minimum	Effective-potential chart	Quotient descent	Not primitive.
Lower branch	Candidate lower channel	Static, lifetime, or thermal verdict	Deleted as primitive.
Metastability	Lifetime class	R_{life} , lifetime comparison, uncertainty envelope	Certificate-mediated.
Instability	Lifetime or static class	Static/lifetime certificate	Certificate-mediated.
Instability crossing	Running route marker	R_H^{scale} plus route certificate	Not primitive.
Bounce route	Decay protocol	Bounce, prefactor/envelope, comparison domain	Route-available or closed.
Thermal transition	Finite-temperature route	Thermal route certificate	Not primitive.
EWBG-style route	Production/preservation protocol	Departure, CP/source, transport, washout, witness	Not primitive.
Richer-scope mechanism	Target update	Carrier/source update and descent certificate	Not same-scope primitive.

B Standing closure certificate fields

Definition B.1 (Vacuum closure certificate). A vacuum standing-closure certificate contains the fields required by its route type:

$$\text{Cert}_{\text{vac}} = (\text{Gauge}, \text{Scheme}, \text{Matching}, \text{Scale}, \text{Route}, \text{Comparison}, \text{Uncertainty}, \text{Witness}).$$

For lifetime claims, the route field contains decay/tunneling and lifetime data. For thermal claims, it contains finite-temperature and transition data. For EWBG-style claims, it contains production, source, transport, and preservation data. For target updates, it includes the carrier/source update and descent map.

Table 6: Standing closure certificate fields

Field	Purpose	Failure if omitted
Gauge	Prevents gauge-fixed representative from becoming endpoint.	Representative promotion.
Scheme	Preserves renormalization-scheme discipline.	Scheme-dependent verdict.
Matching	Relates measured anchors to route inputs.	Target drift or anchor mismatch.
Scale	Routes running/crossing data through R_H^{scale} .	Primitive scale revival.
Route	Supplies static, lifetime, thermal, or EWBG protocol.	Description only.
Comparison	States threshold, time, temperature, or observed-asymmetry target.	No verdict criterion.
Uncertainty	Propagates anchor, perturbative, and route uncertainty.	Rival-basin non-deletion.
Witness	Supplies empirical or structural witness for closure/update.	Unsupported endpoint.

C Route availability versus standing closure ledger

Table 7: Route availability and standing closure ledger

Claim	Availability status	Closure burden	Standing result
Lower branch exists	Representative/marker	Route and verdict certificate	No primitive standing.
RG-improved lower region	Route marker	R_H^{scale} , matching, lifetime or static certificate	Marker only.

Claim	Availability status	Closure burden	Standing result
Bounce calculation submitted	Candidate route	Lifetime certificate and comparison envelope	Available or closed.
Long lifetime claimed	Route-normal if fields supplied	Uncertainty/rival deletion and threshold	Certificate-mediated.
Thermal crossover/transition route	Thermal route	Thermal verdict certificate	Downstream thermal status.
EWBG route	Route target	Departure, CP/source, washout, transport	Route closed only if certified.
BSM vacuum stabilization	Target update	New carrier and descent witness	Richer scope.

D Lower-branch failure register

Table 8: Lower-branch failure register

Primitive attempt	Failed component	Correct route	Verdict
Branch as Higgs identity	Id_H already fixed by C_H .	Representative of route data.	Deleted.
Branch as Higgs scale	Paper 3 deletes primitive scale.	Carrier-indexed scale marker.	Deleted.
Branch as false-vacuum endpoint	Needs lifetime verdict.	R_{life} .	Certificate-mediated.
Branch as thermal history	Different target environment.	R_{therm} .	Route status.
Branch rescued by BSM	Carrier/source update.	$D_{H,+}^{\text{vac}}$.	Target update.
Branch as ZPE/absolute offset	Offset not endpoint standing.	Source/gravity ledger if declared.	Deleted as primitive.

E Thermal and EWBG route-status register

Table 9: Thermal and EWBG route-status register

Route	Necessary components	Same-scope status	Primitive verdict
Thermal crossover	Finite-temperature verdict map	Thermal route status	Not primitive.
First-order EWPT	Barrier/latent heat/bubble or equivalent	Same-scope only if in target; otherwise update	Not primitive.

Route	Necessary components	Same-scope status	Primitive verdict
Bubble nucleation	Thermal route and comparison data	Route status	Not primitive.
Sphaleron/washout	EWBG preservation ledger	Route component	Not primitive.
CP/source route	Source and transport ledger	Route component or target update	Not primitive.
Observed-asymmetry route	Production and preservation certificate	Standing-closed only if certified	Not primitive.

F Claim and nonclaim register

Table 10: Claim and nonclaim register

ID	Statement	Status
V4-T1	Vacuum descriptions are downstream of C_H and ld_H .	Theorem.
V4-T2	Vacuum representatives do not supply primitive Higgs standing.	Theorem.
V4-T3	Same-scope standing-bearing vacuum claims factor through $\text{NF}_{H,\text{vac}}^{\text{stand}}$ and R_H^{vac} .	Theorem.
V4-T4	Lifetime standing closure factors through R_{life} .	Theorem.
V4-T5	Route availability does not imply standing closure; availability-without-closure witness classes exist.	Theorem.
V4-T6	Lower branches have no primitive same-scope survivor.	Main theorem.
V4-T7	Instability crossings are route markers, not primitive endpoints.	Theorem.
V4-T8	Thermal and EWBG-style claims are route statuses or target updates.	Theorem.
V4-T9	Richer-scope vacuum mechanisms require target-update normal form.	Theorem.
V4-T10	$\text{Surv}_{D_H}^{\text{prim}}(\text{Vac}_H) = 0$.	Main theorem.
V4-T11	Every submitted vacuum claim receives layered raw-role, route-status, and standing-status classification.	Theorem.
V4-N1	The paper does not compute a vacuum lifetime.	Nonclaim.
V4-N2	The paper does not select absolute stability, metastability, or instability numerically.	Nonclaim.
V4-N3	The paper does not reject standard effective-potential or metastability calculations.	Nonclaim.
V4-N4	The paper does not rule out BSM vacuum mechanisms.	Nonclaim.
V4-N5	The paper does not independently derive physical-Higgs finite-temperature crossover.	Nonclaim.
V4-N6	The paper does not independently derive EWBG failure or success.	Nonclaim.

ID	Statement	Status
V4-N7	The paper does not reopen Higgs identity or scale standing; both are inherited.	Nonclaim.

G Denial-obligation register

A same-target objection has standing only if it supplies at least one of the following:

1. breaks the inherited carrier package $(D_H, C_H, H_{\text{role}}, R_H, \mathbf{SF}_H)$;
2. breaks the identity endpoint $\text{Id}_H = \text{Occ}_{D_H}(C_H)$;
3. breaks the no-primitive-scale result or R_H^{scale} ;
4. shows that vacuum descriptions can supply primitive standing without representative promotion;
5. breaks the vacuum route normal form R_H^{vac} ;
6. proves $\text{RouteAvail} \Rightarrow \text{StandingClosed}$ without an independent certificate;
7. supplies primitive lower-branch standing that does not reduce to static, lifetime, thermal, or target-update status;
8. supplies a same-scope thermal or EWBG route that changes primitive standing without route or target update;
9. shows a richer-scope mechanism is same-scope while changing no carrier, source, target, or comparison invariant;
10. supplies a faithful same-domain counterexample preserving all inherited arc dependencies;
11. breaks the layered classifier by showing that raw role, route status, and standing status are rival statuses rather than distinct layers;
12. breaks the split between $\mathbf{NF}_{H,\text{vac}}^{\text{route}}$ and $\mathbf{NF}_{H,\text{vac}}^{\text{stand}}$ by showing that an evaluable route closes standing without the certificate field.

Absent one of these results, the objection either changes target, imports hidden selector work, treats a representative as endpoint standing, or refuses AASC vocabulary while retaining the same governance work.

H Source-support manifest

Table 11: Source-support manifest

Support source	Imported role	Local use
Paper 1: Higgs Bridge Carrier	$D_H, C_H, H_{\text{role}}$, local standing filter	Carrier and target dependency.
Paper 2: Higgs Structural Identity	$\text{Id}_H = \text{Occ}(C_H)$	Prevents vacuum descriptions from identity grounding.

Support source	Imported role	Local use
Paper 3: No Primitive Higgs Scale	R_H^{scale} , primitive scale deletion	Handles running, crossing, and instability scale markers.
Kernel paper	Four-role kernel and fixed-domain construction	Background dependency through Paper 1.
Structure of Admissibility	AMetricity, bivalence, standing conservation	Blocks availability-to-standing promotion.
Fixed-Domain Exhaustion	No fifth governance-bearing locus	Supports route-status exhaustion.
Closure by Exhaustion	Same-scope operator closure	Blocks route repair and hidden totalization.
Impossibility Suite	No generators, no transfer carriers, no repairs	Blocks route laundering.
No Primitive Lower Higgs Vacuum	Detailed lower-branch endpoint theorem	Companion route-status and lower-branch precedent.
Arc plan	Six-paper architecture and Paper-4 role	Structural alignment.

I Notation and glossary

Table 12: Notation and glossary

Symbol	Meaning
D_H	Fixed same-scope Higgs-domain target inherited from Paper 1.
C_H	Higgs bridge/fixation carrier.
Id_H	Higgs structural identity endpoint.
R_H^{scale}	Carrier-indexed Higgs scale normal form.
Vac_H	Higgs-vacuum claim class.
R_H^{vac}	Vacuum route normal form.
$\text{NF}_{H,\text{vac}}^{\text{route}}$	Route normalizer for evaluable vacuum routes.
$\text{NF}_{H,\text{vac}}^{\text{stand}}$	Standing normalizer for certificate-closed vacuum routes.
R_{life}	Lifetime route normal form.
RouteAvail	Route availability; form/evaluability status, not standing closure.
StandingClosed	Certificate-closed standing status.
LowerBranch $_H$	Claim that a lower branch supplies Higgs-vacuum standing.
ThermalRoute	Finite-temperature route status.
Standingsmuggling	Illicit conversion of representative or route availability into standing.
Targetupdate	Richer-scope target change.